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SEMICONDUCTOR DEVICE HAVING WORK FUNCTION METAL STACK

Abstract

A device includes gate spacers over a substrate, and a gate structure between the gate spacers. The gate structure includes an interfacial layer, a high-k dielectric layer over the interfacial layer, a tungsten-containing layer over the high-k dielectric layer, and a first metal nitride layer having an underside cupped by the tungsten-containing layer in a cross-sectional view. In the cross-sectional view, the tungsten-containing layer has a U-shaped pattern, the interfacial layer has a linear pattern, and the linear pattern of the interfacial layer extends a length greater than a distance between outermost sides of the U-shaped pattern of the tungsten-containing layer.

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Background/Summary

PRIORITY CLAIM AND CROSS-REFERENCE [0001] This application is a continuation of U.S. patent application Ser. No. 18/588,783, filed Feb. 27, 2024, which is a continuation of U.S. patent application Ser. No. 18/066,203, filed Dec. 14, 2022, now U.S. Pat. No. 11,948,800, issued Apr. 2, 2024, which is a continuation of U.S. patent application Ser. No. 17/549,673, filed Dec. 13, 2021, now U.S. Pat. No. 11,823,908, issued Nov. 21, 2023, which is a continuation of U.S. patent application Ser. No. 16/701,009, filed Dec. 2, 2019, now U.S. Pat. No. 11,201,059, issued Dec. 14, 2021, which is a continuation of U.S. patent application Ser. No. 15/965,635, filed Apr. 27, 2018, now U.S. Pat. No. 10,497,571, issued Dec. 3, 2019, all of which are herein incorporated by reference in their entirety.

BACKGROUND

[0002] The semiconductor integrated circuit (IC) industry has experienced rapid growth. In the course of the IC evolution, functional density (defined as the number of interconnected devices per chip area) has generally increased while geometry size (i.e., the smallest component (or line) that can be created using a fabrication process) has decreased. A scaling down process generally provides benefits by increasing production efficiency and lowering associated costs. But, such scaling down has increased the complexity of processing and manufacturing ICs. For these advances to be realized, similar developments in IC manufacturing are needed.

[0003] To further improve a performance of a semiconductor device, a work function of a metal gate stack of the semiconductor device may be adjusted. In some instances, the adjustment is performed by depositing several work function metal layers having different materials and/or removing a portion of the work function metal layers.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0005] FIG. 1 is a schematic cross-sectional view showing a device in accordance with some embodiments of the present disclosure.

[0006] FIG. 2 is a schematic cross-sectional view showing a device in accordance with other embodiments of the present disclosure.

[0007] FIG. 3 is a schematic cross-sectional view showing a device in accordance with further embodiments of the present disclosure.

[0008] FIG. 4A through FIG. 4I are cross-sectional views of various intermediate stages showing a method of forming a semiconductor device in accordance with some embodiments of the present disclosure.

[0009] FIG. 4C' is a cross-sectional view of intermediate stage showing an implanting operation in accordance with some embodiments of the present disclosure.

[0010] FIG. 4D' is a partially enlarged view of FIG. 4D.

[0011] FIG. 5 is a flowchart showing a method of forming a device in accordance with some embodiments of the present disclosure.

[0012] FIG. 6 is a flowchart showing a method of forming a device in accordance with some embodiments of the present disclosure.

DETAILED DESCRIPTION

[0013] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0014] The spatially relative terms are intended to encompass different orientations of the structure in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0015] A metal gate stack mainly includes a dielectric layer, plural work function metal layers on the dielectric layer, and a gate electrode on the work function metal layers. In pursuit of less device defects, a higher yield and a better device performance of a semiconductor device, various operations may be performed before, during and/or after a metal gate stack formation scheme. For example, one or more work function metal materials may be deposited and patterned over the dielectric layer, thereby forming the metal work function layers of the metal gate stacks. Some impurities may be introduced into one or more metal work function layers to further adjust a work function of the work function metal layers. Introducing the impurities may be performed by depositing a work function adjusting layer on the metal work function layer, annealing the substrate to diffuse the impurities of the work function adjusting layer into work function metal layers underlying the work function adjusting layer, and removing the work function adjusting layer. The work function metal layer directly underlying the work function adjusting layer may be used as a protective layer, when one or more etching operations are performed to remove the work function adjusting layer, so as to prevent damages of the underlying features (e.g. the other work function metal layer and/or the dielectric layer). Optionally, introducing the impurities is performed by implanting the impurities into one or more work function metal layers.

[0016] However, the work function metal layer directly underlying the work function adjusting layer or directly contacting the implanting source has a high risk of a structure loss (or damage) resulted from the etching operations or the implanting operation. When the damaged work function metal layer remains in the metal gate stack, serious device defects (e.g. a N/P boundary concern of the semiconductor device, impacts on a threshold voltage ($V_{sub.t}$) and yielding of the semiconductor device) may occur. Furthermore, the damage of the work function metal layer may become worse when the manufacturing process of the metal gate stack goes further.

[0017] Embodiments of the present disclosure are directed to providing a device and a method of

forming the same, in which the device includes plural work function metal layers. One aspect of the embodiments of the present disclosure is to remove the damaged work function metal layer and re-deposit a new work function metal layer. In some embodiments, the re-deposited work function metal layer has similar composition to the removed one. In at least one embodiment, during removing the damaged work function metal layer, a tungsten (W)-containing residue remains, and the W-containing residue may be referred to as an evidence that the removing operation and the re-deposition operation has been performed. The method of forming the device of the present disclosure provides various benefits, such as no need of an additional mask, being compatible to a current semiconductor manufacturing process, an enough strength of the re-deposited work function layer for subsequent patterning processes, and less device defects, so as to form the semiconductor device having a higher yielding and a better performance.

[0018] In some embodiments, the semiconductor device mentioned above may include a metal-oxide semiconductor field effect transistor (MOSFET), a fin field effect transistor (FinFET), a tunnel field-effect transistor (T-FET), a device having nanowire channels, a planar semiconductor device and the like. The scope of the present disclosure may be applied to a variety of the semiconductor devices, and may not be limited to the examples shown in the present disclosure.

[0019] Referring to FIG. 1, FIG. 1 is a schematic cross-sectional view showing a device **100** in accordance with some embodiments of the present disclosure. As shown in FIG. 1, the device **100** includes a substrate **102**, an interlayer dielectric (ILD) layer **104**, gate spacers **106** and a metal gate stack **100'**. The ILD layer **104** is disposed on the substrate **102**, and the gate spacers **106** and the metal gate stack **100'** are disposed in the ILD layer **104**. The gate spacers **106** are disposed on sidewalls of the metal gate stack **100'**. The metal gate stack **100'** includes an interfacial layer **110**, a dielectric layer **120**, a first work function metal layer **130**, an intermediate layer **140**, a second work function metal layer **150** and a metal gate electrode **160**. The interfacial layer **110** is disposed over the substrate **102** and at the bottom of the metal gate stack **100'**. The dielectric layer **120** is disposed over the interfacial layer **110**. The first work function metal layer **130** is disposed over the dielectric layer **120**. The intermediate layer **140** is disposed over the first work function metal layer **130**. The second work function metal layer **150** is disposed over the intermediate layer **140**. The metal gate electrode **160** is disposed over the second work function metal layer **150**. The dielectric layer **120**, the first work function metal layer **130**, the intermediate layer **140** and the second work function metal layer **150** peripherally surround the metal gate electrode **160**. In some embodiments, the device **100** may have 3, 4, 5 or 6 work function (WF) layers. In addition, both p-type and n-type WF layers may be included in an n-type FET; and, both p-type and n-type WF layers may be included in a p-type FET.

[0020] In some embodiments, the substrate **102** may be a silicon substrate. In other embodiments, the substrate **102** may alternatively include germanium, other III-V elements and/or their compounds such as silicon germanium, gallium arsenic, or any other suitable semiconductor materials. In a further embodiment, the substrate **102** may include other features such as various doped regions such as p-wells or n-wells, a buried layer, and/or an epitaxy layer. Furthermore, the substrate **102** may be a semiconductor on insulator such as silicon on insulator (SOI). In other embodiments, the semiconductor substrate **102** may include a doped epi layer, a gradient semiconductor layer, and/or may further include a semiconductor layer overlying another semiconductor layer of a different type such as a silicon layer on a silicon germanium layer. In other examples, a compound semiconductor substrate may include a multilayer silicon structure or a silicon substrate may include a multilayer compound semiconductor structure.

[0021] In some embodiments, the ILD layer **104** may include flowable oxide. The ILD layer **104** may also be a spin-on glass. For example, the ILD layer **104** may include phospho-silicate glass (PSG), boro-silicate glass (BSG), boron-doped phospho-silicate glass (BPSG), tetraethyl orthosilicate (TEOS) oxide, TIN, SiOC, or other dielectric materials.

[0022] In some embodiments, the gate spacers **106** are formed of silicon oxide, silicon nitride,

silicon carbon nitride, or the like. Furthermore, the gate spacers **106** may have a multi-layer structure, for example, with a silicon nitride layer over a silicon oxide layer. In some embodiments, the interfacial layer **110** includes an oxide layer such a silicon oxide layer.

[0023] In some embodiments, the dielectric layer **120** is a high-k dielectric layer. In one example, a material of the high-k dielectric layer may include but is not limited to hafnium oxide ($\text{HfO}_{x\cdot y}$), LaO , AlO , ZrO , TiO , $\text{Ta}_{2\cdot 5}\text{O}_3$, $\text{Y}_{2\cdot 3}\text{O}_3$, SrTiO_3 (STO), BaTiO_3 (BTO), BaZrO , HfZrO , HfZrON , HfLaO , HfSiO , HfSiON , LaSiO , AlSiO , HfTaO , HfTiO , $(\text{Ba}, \text{Sr})\text{TiO}_3$ (BST), $\text{Al}_{2\cdot 3}\text{O}_3$, $\text{Si}_{3\cdot 4}\text{N}_4$, oxynitrides, or other suitable materials, where $x > 0$.

[0024] In some embodiments, the first work function metal layer **130** and the second work function metal layer **150** includes different work function metals. In some embodiments, the first work function metal layer **130** includes TiN , $\text{TiNO}_{\cdot y}$, $\text{TaNO}_{\cdot y}$, Al-doped TiN , or a combination thereof, in which $y > 0$. In other examples, the second work function metal layer **150** may include TaN , W or a nitride compound containing Ti and Ta . In some embodiments, a thickness of each of the first work function metal layer **130** and the second work function metal layer **150** is in a range from about 5 Å to about 25 Å. In some embodiments, the second work function metal layer **150** is in a range from about 5 Å to about 15 Å. A target work function depends on the thickness of the first and second work function metal layers **130** and **150**, and can be modified according to requirements of a semiconductor device.

[0025] In some embodiments, the intermediate layer **140** may include a compound different from the first work function metal layer **130**, such as a nitride compound of aluminum, tungsten, tantalum, or titanium. In some embodiments, the intermediate layer **140** includes at least one of a binary, ternary, and quaternary tungsten compound. For example, the compound of tungsten may include, but is not limited to, tungsten chloride, tungsten oxide, or a combination thereof which are formed by a residue or a byproduct of a removed work function metal layer. In some embodiments, the intermediate layer **140** may further include a compound of W and TIN , TaN , $\text{TiNO}_{\cdot x}$ and/or $\text{TaNO}_{\cdot x}$ which may come from a portion of the removed work function metal layer, and $x > 0$. In some embodiments, the intermediate layer **140** has a thickness in a range from about 3 Å to about 10 Å. A greater thickness increases the difficulty of subsequent filling process, in some instances. A smaller thickness would damage the underlying layers, in some instances. Details will be described later. In some embodiments, the intermediate layer **140** is detectable under ZC-mode of an electron microscope image, such as a transmission electron microscope (TEM), in which the intermediate layer **140** is relatively thin and is presented in an interface between two work function metal layers (e.g. between the first work function metal layer **130** and the second work function metal layer **150**). In some embodiments, the intermediate layer **140** includes a same material as the first work function metal layer **130**. In some embodiments, the metal gate electrode **160** may include a conductive material, such as aluminum, tungsten, copper.

[0026] Referring to FIG. 2, FIG. 2 is a schematic cross-sectional view showing a device **200** in accordance with other embodiments of the present disclosure. As shown in FIG. 2, the device **200** includes a substrate **202**, an ILD layer **204**, gate spacers **206** and a metal gate stack **200'**. The substrate **202**, the ILD layer **204** and the gate spacers **206** are similar to the substrate **102**, the ILD layer **104** and the gate spacers **106**. The metal gate stack **200'** includes an interfacial layer **210**, a dielectric layer **220**, two first work function metal layers **230** and **232**, an intermediate layer **240**, a second work function metal layer **250**, a third work metal function **252** and a metal gate electrode **260**. The interfacial layer **210**, the dielectric layer **220**, the intermediate layer **240**, the second work function metal layer **250** and the metal gate electrode **260** are similar to the interfacial layer **110**, the dielectric layer **120**, the intermediate layer **140**, the second work function metal layer **150** and the metal gate electrode **160**. In some embodiments, each of the first work function metal layers **230** and **232** may have the similar composition to the first work function metal layer **130** described in FIG. 1, and the first work function metal layers **230** may be same as or different from the first

work function metal layers **232**. In some embodiments, the third work function metal layer **252** include a p-type work function metal similar to the first work function metal layer **130** or the second work function metal layer **150** described in FIG. **1**. In other embodiments, the third work function metal layer includes an n-type work function metal such as aluminum (Al) and a compound of aluminum, e.g. aluminum oxide, aluminum nitride and any other suitable compounds. In some embodiments, a thickness of each of the first work function metal layers **230** and **232**, the second work function metal layer **250** and the third work function metal layers **252** is in a range from about 5 Å to about 25 Å. In some embodiments, the second work function metal layer **250** is in a range from about 5 Å to about 15 Å.

[0027] From the embodiments shown in FIG. **2**, the relative location (among the other work function metal layers) of the intermediate layer **240** is not fixed, and the relative location of the intermediate layer **240** may be adjusted depending on the configuration or the design of the semiconductor device. The examples provided in the figures are not intended to limit the relative location of the intermediate layer, but only shown for clear understanding of the present disclosure. A skilled person in the art may easily adjust the numbers of the first and third work function metal layers according to the requirements or the design of the device.

[0028] Referring to FIG. **3**, FIG. **3** is a schematic cross-sectional view showing a device **300** in accordance with further embodiments of the present disclosure. In FIG. **3**, the device **300** includes a substrate **302**, an interlayer dielectric layer **304**, gate spacers **306**, a metal gate stack **300A** and a metal gate stack **300B**. The substrate **302**, the ILD layer **304** and the gate spacers **306** are similar to the substrate **102**, the ILD layer **104** and the gate spacers **106**, in which a shallow trench isolation (STI) **308** is disposed in the substrate **302** and between the metal gate stack **300A** and the metal gate stack **300B**. Interfacial layers **310A** and **310B**, dielectric layers **320A** and **320B**, first work function metal layers **330A** and **330B**, intermediate layers **340A** and **340B**, second work function metal layers **350A** and **350B** and metal gate electrodes **370A** and **370B** of the metal gate stacks **300A** and **300B** in FIG. **3** are respectively similar to the interfacial layer **110**, the dielectric layer **120**, the first work function metal layer **130**, the intermediate layer **140**, the second work function metal layer **150** and the metal gate electrode **160** of the metal gate stack **100'** of FIG. **1**. A third work function metal layer **360A** of the metal gate stack **300A** includes the n-type work function metal, for example, aluminum (Al) and a compound of aluminum, e.g. aluminum oxide, aluminum nitride and any other suitable compounds. A third work function metal layer **360B** of the metal gate stack **300B** includes the p-type work function metal, for example, TaN, TiN, TiNO.sub.y, TaNO.sub.y, Al-doped TiN, W, or a combination thereof, and y>0. Furthermore, a fourth work function metal layer **362** may be disposed over the third work function metal layer **360B** of the metal gate stack **300B**, and the fourth work function metal layer **362** may be formed by the similar material as the third work function metal layer **360A** of the metal gate stack **300A**.

[0029] In some embodiments, the metal gate stack **300A** may be used to form a n-type metal device, for example, a NMOS device. In other embodiments, the metal gate stack **300B** may be used to form a p-type device, for example, a PMOS device. In a further embodiment, the device **300** may be a CMOS device (e.g. a statistic random access memory (SRAM)).

[0030] The metal gate stack **300B** has a thicker (or greater amounts of) p-type work function metal (PWFM) layers than the metal gate stack **300A**. However, the number of the PWFM layers shown in FIG. **3** is only for convenience of descriptions, and is not intended to limit the scope of the present disclosure. In a further example, the metal gate stack **300B** may have the structure shown in FIG. **2**. In other embodiments, the metal gate stack **300B** may include a first PWFM layer **330B**, a second PWFM layer **350B** and two third PWFM layers **360B**, and the intermediate layer **340B** disposed between the first PWFM layer **330B** and the second PWFM layer **350B**. In another embodiment, the metal gate stack **300B** may include a first PWFM layer **330B**, a second PWFM layer **350B** and three third PWFM layers **360B**, and the intermediate layer **340B** disposed between the first PWFM layer **330B** and the second PWFM layer **350B**. The embodiment of FIG. **3** shows

the n-type work function metal (NWFM) layer **362** disposed on the third PWFWM layers **360B** of the metal gate stack **300B**, however, it is not necessary for the p-type metal gate stack to have the NWFM layer. In the embodiments where the metal gate stack **300B** includes two or three third PWFWM layers **360B**, the metal gate stack **300A** may further include a third PWFWM layer between the second PWFWM layer **350A** and the NWFM layer **360A**, or the metal gate stack **300A** may include plural first PWFWM layers **330A**.

[0031] Referring to FIG. **4A** through FIG. **4I**, FIG. **4A** through FIG. **4I** are cross-sectional views of various intermediate stages showing a method of forming a device in accordance with some embodiments of the present disclosure. As shown in FIG. **4A**, a substrate **402** is provided, and a STI **408** is formed in the substrate **402**. Typically, dummy gate stacks (not shown) are formed on the substrate **402** and isolated by the STI **408**, in which the dummy gate stacks are formed by a series of deposition processes and patterning processes. The dummy gate stacks are located at the position of trenches **401** and **403** formed later, as shown in FIG. **4A**. Gate spacers **406** are formed on sidewalls of each of the dummy gate stacks, in which the formation of the gate spacers **406** may include a dielectric deposition process, an anisotropic etching process and a planarization process. Then, an inter-layer dielectric (ILD) layer **404** may be formed on the substrate **402**. For example, the ILD layer **404** may be formed using flowable chemical vapor deposition (FCVD) or spin-on coating. Next, removing the dummy gate stacks, thereby forming the trenches **401** and **403** in the ILD layer **404**, as shown in FIG. **4A**. Then, an interfacial layer **410** and a dielectric layer **420** are formed in each of the trenches **401** and **403**. In some embodiments, the interfacial layer **410** may be formed through thermal oxidation process, a chemical oxidation, or a deposition process. In some embodiments, the dielectric layer **420** may be formed by an atomic layer deposition (ALD) process, a chemical vapor deposition (CVD) process, or the like. In some embodiments, the substrate **402**, the ILD layer **404**, the gate spacers **406**, the interfacial layer **410** and the dielectric layer **420** are respectively same as the substrate **102**, the ILD layer **104**, the gate spacers **106**, the interfacial layer **110** and the dielectric layer **120** in FIG. **1**. The formation of the intermediate structure of FIG. **4A** is known to a person skilled in the art, and may not be further described herein.

[0032] Next, as shown in FIG. **4B**, a first work function metal layer **430** is deposited over the dielectric layer **420**, followed by depositing a dummy layer **440** over the first work function metal layer **430**. In some embodiments, the first work function metal layer **430** has an etch selectivity different from that of the dummy layer **440**. The first work function metal layer **430** and the dummy layer **440** include different p-type work function metal materials. In some embodiments, the material of the first work function metal layer **430** includes TiN, TiNO.sub.y, TaNO.sub.y, Al-doped TiN, or a combination thereof, and y>0. In some embodiments, the material of the dummy layer **440** includes TaN, W or a nitride compound containing Ti and Ta.

[0033] Then, an impurity is introduced into the first work function metal layer **430**, such that a work function of the first work function metal layer **430** may be adjusted. In the embodiment of FIG. **4C**, introducing the impurity is performed by depositing a work function adjusting layer **442** over the dummy layer **440**, followed by annealing the work function adjusting layer **442** to diffuse the impurity into at least one of the dummy layer **440** and the first work function metal layer **430**, such as an operation **444** shown in FIG. **4C**. The impurity may be further introduced into the dielectric layer **420**. The impurity is selected based on a predetermined work function, and there is no limitation of a species of the impurity in the present disclosure. After the annealing operation **444**, the work function adjusting layer **442** is etched such that a top portion of the dummy layer **440** is etched, as shown in FIG. **4D** and FIG. **4D'**. In some embodiments, etching the work function metal adjusting layer is performed using a wet etching operation. When the work function adjusting layer **442** is etched, the dummy layer **440** is used as a protective layer (i.e. an etch stop layer) and directly contacts the etchant of the etching operation, causing damages or a loss of the dummy layer **440**. For example, the top portion of the dummy layer **440** becomes uneven or rough in the enlarged FIG. **4D'**.

[0034] In some embodiments, the operations of depositing, annealing and etching the work function metal adjusting layer **442**, may be repeated for several times, so as to achieve a desired characteristic. The characteristic depends on a type of the semiconductor device, a target performance of the semiconductor device or any other conditions, and thus the present disclosure is not limited to any specific method for adjusting the work function. Any techniques known by a skilled person in the art to adjust the work function of the semiconductor device can be applied here.

[0035] In other embodiments, introducing the impurity is performed by implanting the impurity into the first work function metal layer **430**, such as an operation **446** shown in FIG. **4C'**. After the implanting operation **446**, the top portion of the dummy **440** is also likely to be damaged, as shown in the enlarged FIG. **4D'**.

[0036] Typically, the damaged dummy layer **440** remains as one of the work function metal layers, and another work function metal layer(s) may then be directly deposited on the dummy layer **440**. However, the formed semiconductor device may have a poor performance due to the damaged dummy layer **440**. Therefore, as shown in FIG. **4E**, the dummy layer **440** is removed, such that a W-containing residue **440'** is formed over the first work function metal layer **430**. In some embodiments, removing the dummy layer **440** is performed by a wet etching process using an etchant. In some embodiments, the etchant may include but is not limited to tungsten fluoride, tungsten chloride, a combination thereof, or any other suitable etchants. In some embodiments, the removal of the dummy layer **440** is performed by a dry etching operation. In some embodiments, the dummy layer **440** is partially removed. In some embodiments, the dummy layer **440** is completely removed. In some embodiments, the method further includes a cleaning operation to remove some byproducts formed during the removal of the dummy layer **440** while the W-containing residue **440'** remains.

[0037] In some embodiments, the W-containing residue **440'** is formed by a reaction between the etchant and the dummy layer **440**. In some embodiments, the W-containing residue **440'** includes a compound different from the first work function metal layer **430**, such as a nitride compound of aluminum, tungsten, tantalum, or titanium. In some embodiments, the W-containing residue **440'** includes at least one of a binary, ternary, and quaternary tungsten compound. In this embodiment, the W-containing residue **440'** may include the compound of tungsten including but is not limited to, tungsten chloride, tungsten oxide, or a combination thereof. In a further embodiment, the W-containing residue **440'** may be a remaining portion of the dummy layer **440**, and thus the W-containing residue **440'** may further include a compound of W and TIN, TaN, TiNO.sub.x and/or TaNO.sub.x, or a combination thereof, and $x > 0$. In some embodiments, the W-containing residue **440'** has a thickness in a range from about 3 Å to about 10 Å. A greater thickness increases the difficulty of subsequent filling process, in some instances. A smaller thickness would damage the underlying layers, in some instances.

[0038] Then, as shown in FIG. **4F**, a second work function metal layer **450** is deposited over the first work function metal layer **430** and the W-containing residue **440'**. In some embodiments, depositing the second work function metal layer **450** is performed such that the W-containing residue **440'** remains between the first work function metal layer **430** and the second work function metal layer **450** after depositing the second work function metal layer **450**. In one embodiment, the second work function metal layer **450** includes a p-type work function metal, and the second work function metal layer **450** and the dummy layer **440** may include the same material. In other words, the formation of the second work function metal layer **450** may be referred to re-deposition of the dummy layer **440**. In other embodiments, the second work function metal layer **450** may be formed by different materials from the dummy layer **440**.

[0039] Still referring to FIG. **4F**, a third work function metal layer **460** is deposited over the second work function metal layer **450** in some embodiments. The third work function metal layer **460** includes TaN, TIN, TiNO.sub.y, TaNO.sub.y, Al-doped TIN, W, or a combination thereof, and $y > 0$.

There is no limitation for the material of the third work function metal layer **460**, for example, the third work function metal layer **460** is same as the first work function metal layer **430** in some embodiments. In other embodiments, the third work function metal layer **460** is same as the second work function metal layer **450**. In a further embodiment, the third work function metal layer **460** is different from the first work function metal layer **430** and the second work function metal layer **450**. In some embodiments, each of the first work function metal layer **430**, the second work function metal layer **450** and the third work function metal layer **460** has a thickness in a range from about 5 Å to about 25 Å. In some embodiments, a thickness of the second work function metal layer **450** is in a range from about 5 Å to about 15 Å. When the thickness of the second work function metal layer **450** is out of the above range, the threshold voltage varies.

[0040] Next, as shown in FIG. **4G** and FIG. **4H**, a portion of the third work function metal layer **460** in the trench **403** and on the ILD layer **404** is removed, followed by depositing a fourth work function metal layer **470** in the trench **401** and trench **403** and on the ILD layer **404**. During removing the portion of the third work function metal layer **460**, the redeposited second work function metal layer **450** also acts as a barrier layer/protection layer. In some embodiments, the fourth work function metal layer **470** includes an n-type work function metal, such as aluminum (Al) and a compound of aluminum, e.g. aluminum oxide, aluminum nitride and any other suitable compounds. The removal of the portion of the third work function metal layer **460** may be performed by a N/P patterning process using a mask (not shown) formed over the trench **401**. In the embodiments where the fourth work function metal layer **470** is not formed in the trench **401**, the mask is also used in the deposition of the fourth work function metal layer **470**. The N/P patterning process is known by a skilled person in the art and may not be further described herein. In some embodiments, the formation of the first work function metal layer **430**, the dummy layer **440**, the second work function metal layer **450**, the third work function metal layer **460** and the fourth work function metal layer **470** includes a deposition process and a patterning process. In some embodiments, the deposition process includes but is not limited to chemical vapor deposition (CVD), atomic layer deposition (ALD), physical vapor deposition (PVD), or any other suitable processes.

[0041] Still referring to FIG. **4H**, a conductive material **480** is filled into the trenches **401** and **403**. The conductive material **480** includes, for examples, aluminum, tungsten, copper. Then, as shown in FIG. **4I**, a portion of the first work function metal layer **430**, the W-containing residue **440'**, the second work function metal layer **450**, the third work function metal layer **460**, the fourth work function metal layer **470** and the conductive material **480** on the ILD layer **404** are removed, and a device **400** is formed. The device **400** includes an n-type device **400A** and a p-type device **400B**. The removal of the above portions may be performed by a planarization operation, such as chemical mechanical planarization (CMP).

[0042] FIG. **5** is a flowchart showing a method **500** of forming a device in accordance with some embodiments of the present disclosure. FIG. **4A** through FIG. **4I** are incorporated as a reference for clear understanding.

[0043] At operation **510**, a dielectric layer **420** is deposited over a substrate **402**, as shown in FIG. **4A**. At operation **520**, a first work function metal layer **430** is deposited over the dielectric layer **420**, as shown in FIG. **4B**. Then, at operation **530**, a dummy layer **440** is deposited over the first work function metal layer **430**, as shown in FIG. **4B**. Next, at operation **540**, an impurity is introduced into the first work function metal layer **430**, as shown in FIG. **4C**, FIG. **4C'**, FIG. **4D** and FIG. **4D'**. At operation **550**, the dummy layer **440** is removed, as shown in FIG. **4E**. At operation **560**, a second work function metal layer **450** is deposited over the first work function metal layer **430**, as shown in FIG. **4F**. The method may further include depositing a third work function metal layer **460**, as shown in FIG. **4F**. A portion of the third work function layer **460** is removed, and a fourth work function metal layer **470** and a conductive material **480** are sequentially deposited, as shown in FIG. **4G** and FIG. **4H**. Then, portions of the first, second, third

and fourth work function metal layers **430**, **450**, **460** and **470**, the W-containing residue **440'** and the conductive material **480** are planarized, thereby forming the device **400** shown in FIG. **4I**. [0044] FIG. **6** is a flowchart showing a method **600** of forming a device in accordance with some embodiments of the present disclosure. FIG. **4A** through FIG. **4I** are incorporated as a reference for clear understanding.

[0045] At operation **610**, a dielectric layer **420** is deposited over a substrate **402**, as shown in FIG. **4A**. At operation **620**, a first work function metal layer **430** is deposited over the dielectric layer **420**, as shown in FIG. **4B**. Then, at operation **630**, a dummy layer **440** is deposited over the first work function metal layer **430**, as shown in FIG. **4B**. At operation **640**, the dummy layer **440** is removed using an etchant, such that a W-containing residue **440'** is formed over the first work function metal layer **430** after removing the dummy layer **440**, as shown in FIG. **4C** through FIG. **4E**. At operation **650**, a second work function metal layer **450** is deposited over the W-containing residue **440'**, as shown in FIG. **4F**. The method may further include depositing a third work function metal layer **460**, as shown in FIG. **4F**. A portion of the third work function layer **460** is removed, and a fourth work function metal layer **470** and a conductive material **480** are sequentially deposited, as shown in FIG. **4G** and FIG. **4H**. Then, portions of the first, second, third and fourth work function metal layer **430**, **450**, **460** and **470**, the W-containing residue **440'** and the conductive material **480** are planarized, thereby forming the device **400** shown in FIG. **4I**.

[0046] The method of forming the device is provided in the present disclosure. The method mainly includes removing the damaged dummy layer and re-depositing another new layer (i.e. the second work function metal layer) that is similar to (e.g. having the same composition) the dummy layer. Therefore, the device formed by the method may include the W-containing layer, which is a residue layer of the dummy layer. The method has benefits such as no need of an additional mask, being compatible to the current semiconductor manufacturing process, less device defects (e.g. N/P boundary concern), so as to form the semiconductor device having a higher yield and a better performance.

[0047] According to some embodiments, a method includes removing a dummy gate to form a gate trench. A gate dielectric layer is deposited over a bottom and sidewalls of the gate trench. A first work function metal layer is deposited over the gate dielectric layer. A dummy layer is deposited over the first work function metal layer. An impurity is introduced into the dummy layer and the first work function metal layer after the dummy layer is deposited. The dummy layer is removed after the impurity is introduced into the dummy layer and the first work function metal layer. The gate trench is filled with a conductive material after the dummy layer is removed.

[0048] According to some embodiments, a method includes removing a dummy gate to form a gate trench. A gate dielectric layer is deposited over a bottom and sidewalls of the gate trench. A first work function metal layer is deposited over the gate dielectric layer. A dummy layer is deposited over the first work function metal layer. A work function adjusting layer is deposited over the dummy layer. The work function adjusting layer is annealed. The work function adjusting layer is removed after the work function adjusting layer is annealed. The dummy layer is removed after the work function adjusting layer is removed. The gate trench is filled with a conductive material after the dummy layer is removed.

[0049] According to some embodiments, a device includes a gate electrode, a first work function metal layer, a gate dielectric layer, and a tungsten-containing material. The first work function metal layer wraps around the gate electrode. The gate dielectric layer wraps around the first work function metal layer. The tungsten-containing material is between the gate electrode and the first work function metal layer and is spaced apart from the gate electrode.

[0050] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages

of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A device, comprising: gate spacers over a substrate; and a gate structure disposed between the gate spacers, the gate structure comprising: an interfacial layer; a high-k dielectric layer over the interfacial layer; a tungsten-containing layer over the high-k dielectric layer; and a first metal nitride layer having an underside cupped by the tungsten-containing layer in a cross-sectional view, wherein in the cross-sectional view, the tungsten-containing layer has a U-shaped pattern, the interfacial layer has a linear pattern, wherein the linear pattern of the interfacial layer extends a length greater than a distance between outermost sides of the U-shaped pattern of the tungsten-containing layer.
2. The device of claim 1, wherein the tungsten-containing layer has a thickness less than a thickness of the interfacial layer.
3. The device of claim 1, wherein the first metal nitride layer comprises TaN or a nitride compound containing Ti and Ta.
4. The device of claim 1, further comprising: a second metal nitride layer cupping an underside of the tungsten-containing layer.
5. The device of claim 4, wherein the second metal nitride layer has a thickness greater a thickness of the tungsten-containing layer.
6. The device of claim 4, wherein the second metal nitride layer interfaces the underside of the tungsten-containing layer.
7. The device of claim 4, wherein the tungsten-containing layer interfaces the underside of the first metal nitride layer.
8. The device of claim 1, further comprising: a tungsten plug having an underside cupped by the first metal nitride layer.
9. The device of claim 8, wherein the tungsten plug is spaced apart from the tungsten-containing layer at least by the first metal nitride layer.
10. A device, comprising: gate spacers over a substrate; and a gate structure between the gate spacers, the gate structure comprising: a gate dielectric layer; a tungsten-containing layer having a U-shape and cupped by the gate dielectric layer in a cross-sectional view; a first metal nitride layer having a U-shaped and cupped by the tungsten-containing layer in the cross-sectional view; and a tungsten plug cupped by the first metal nitride layer and having a different shape than the U-shape of the tungsten-containing layer.
11. The device of claim 10, wherein the tungsten plug is a linear plug extending in a direction perpendicular to the substrate in the cross-sectional view.
12. The device of claim 10, wherein the tungsten-containing layer is a binary tungsten compound, a ternary tungsten compound, or a quaternary tungsten compound.
13. The device of claim 10, wherein the first metal nitride layer comprises TaN or a nitride compound containing Ti and Ta.
14. The device of claim 10, further comprising: a second metal nitride layer, wherein the tungsten-containing layer has an inner surface interfacing the first metal nitride layer, and an outer surface interfacing the second metal nitride layer.
15. The device of claim 14, wherein the second metal nitride layer is thicker than the tungsten-containing layer.
16. The device of claim 14, wherein the second metal nitride layer comprises a different composition than the first metal nitride layer.

- 17.** A device, comprising: a substrate; gate spacers over the substrate; and a gate structure between the gate spacers, the gate structure comprising: a gate dielectric layer; a first metal nitride layer over the gate dielectric layer; and a tungsten-containing layer over the first metal nitride layer and thinner than the first metal nitride layer.
- 18.** The device of claim 17, further comprising: a second metal nitride layer over the tungsten-containing layer and thicker than the tungsten-containing layer.
- 19.** The device of claim 18, wherein the second metal nitride layer comprises TaN or a nitride compound containing Ti and Ta.
- 20.** The device of claim 17, wherein the first metal nitride layer comprises TiN, TiNO.sub.y, TaNO.sub.y, or Al-doped TiN.
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